MOHIT MEHTA

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Passionate about systems, applied ml, systems for ml, ml infra, data infrastructure and software engineering practices.

EXPERIENCE

- **Staff SWE** - Data / ML, Audio & Platform Services - **Nielsen** — Remote · Aug'24 - present

  Part of the data preparation, audio, platform and data factories vertical in Data acquisition org at Nielsen. Working on data ingestion, internal crediting, volumetric analysis and comparisons, feature preparation, audio fingerprint extraction and a self service data/ml platform.

- **Principal SWE** - ML / Core - **Noodle AI** — Remote · Oct'23 - Jan'24 · break

  Building noodle's ml infrastructure, data & compute infra and collaborating with ml research, applied data science org developing demand-supply forecasting and planning models for supply chain planning and building a self-service platform with focus on pipeline topology, performance.

- **Senior SDE** - Demand / Data - **Zomato** — New Delhi · 2019 - '20 · break · 2023

  - Senior SDE - Platform - Data Platform, Service, Messaging — Apr'23 - Aug'23
  - SDE II - Marketplace, Maps & Geo - Demand Experimentation & Serviceability Infrastructure — Apr'19 - Mar'20

  Worked on teams Demand, Discovery & Serviceability on problems around merchant serviceability - experiments on point in-time geo-spatial serviceability to optimize for rider utilization & demand - throttling and forecasting - enabling dynamic serviceability for partners.
  Worked with team building a common platform for support - chats, agent channels, chatroom agent allocation - QoS / progressive load shedding, cloud-calling, ticketing, business bots, proactive support & bot handover for entities in hyperlocal, dining & procurement business.

- **Senior Software Engineer** - Search / AI Infra - **Unbxd Inc. / Netcore Cloud** — Remote · Dec'20 - Dec'22

  - Senior Software Engineer - AI Infrastructure / Platform, Applied AI - Query, Personalization, RecSys
  - Software Engineer - Search Infrastructure, Knowledge

  Worked on teams Platform and Self Serve on building platform services for self-service AI platform (serving 220+ active saas clients with different reference model architectures for entity extraction, item similarity, search personalization [serving recommendation/re-ranking]), recommendations & personalization infrastructure, search platform, feature store, merchandising - extraction platform and knowledge graph.

- **Software Engineer** - Platform - **Saxo Capital Markets** — New Delhi · Jul'17 - Apr'19

  - SE, Senior Associate - Saxo Group - Services
  - SE, Associate - Saxo Capital Markets - Pricing / Data Infrastructure

  Worked with teams under equity research platform and client experience on building platform services for Pricing infrastructure, data feed adapters, strategy test & instruments' research tooling, Onboarding, SaxoTraderGo, client management and vendor services.

- **Software Engineer** - Search / Relevance - **HT Media** — New Delhi · Jul'15 - Jun'17

  Worked with teams related to Recruiter platform, connections and employee referral platform, referral/connection match relevance analytics infrastructure / graph analytical modeling pipeline and shine learning platform.

EDUCATION

- BE - Computer Science - EE/CS - Delhi Technological University, formerly Delhi College of Engineering — New Delhi · 2011 - '15
- MS - Computer Science - Computing Systems & Machine Learning - College of Computing, Georgia Institute of Technology — Part-time · 2020 - '21, 2026 - cont
- Independent - Mathematics and Computer Science - deeplearning.ai, stanford online, uw — Part-time · 2017 - '20, 2024 - cont

PROFICIENCY

- Programming - Java, Python, Go, Scala, js/ts/node, c++
- Data storage & indexing - mysql, postgres, redis, mongodb, cassandra, solr, dgraph, neo4j, dynamodb
- Data Infrastructure - presto, flink, pinot, druid, kdb+, redash, argo, airflow, storm, spark, giraph, hive, snowflake, aws - emr, sagemaker, gcp - bigquery
- Data analysis & Machine learning - exploratory analysis, nlp, DNN architectures, autoML, packages - pandas, sklearn, nltk, pytorch, tensorflow, keras, hf transformers, opencv, flair
- Machine learning Infrastructure - kubeflow, katib, kfp, kf-serving, mlflow, vertex ai, langchain, onxx, openmpi, feast, cudnn, model parallelism (TP/PP), SHAP, tensorrt
- Developer Infrastructure - k8s, helm, flux, prometheus, elk, aws - ecs, eks, sqs, lambda, athena, periscope, gcp - gke, dagster, airflow, dbt, gx, datahub, databricks, pachyderm, ray
- Service communication - kafka, rabbitmq, mqtt, nginx, zookeeper, grpc, envoy, istio, kuma-dp
- Development - *nix tools, eBPF, LSM, frameworks - gin, go-kit, flask, fastapi, django, qPython, node.js, hapi.js, spring-boot, pippo, vert.x, react, android / flutter, osrm

INTERESTS

- Technical Interest areas - Distributed Systems, Scalability, Software Design, Data/ML infrastructure, ml systems, foundation models, optimization techniques, systems for ML/HPC.
- Online Presence - social - linkedin - mohitmehta11, forums - topcoder/stackexchange - zapper123, collegiate - ACM ICPC Asia Regionals 2013 - team - stimuli - regionals participation.
- Personal traits align with an organized, pragmatic, level-headed and humble team player. Leadership traits align with building an all-hands involved empowering and contributing environment.

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NOTABLE CONTRIBUTIONS

- Upstream features and downstream pre-aggregates availability for temporal, hierarchical reconciled modeling — ML Infrastructure · Noodle.ai
  - Owned and built the workflows for data availability, feature availability for forecasting models comprising requirements of features at different hierarchies of a product or a supply chain graph node for modeling a composite set of statistical models at each hierarchy.
  - Built the workflows capturing and aggregating the forecasts at each level and detailing summaries, products of impact/at risk and their forecasts, variance of weights to independent models and contributions of features and their driving data for different forecast target dates.

  Predominant scope - system architecture, ml platform engineering, orchestration, analytics infrastructure, ensemble methods, time series forecasting.
  Technologies used - python, dagster, airflow, kubeflow, druid, snowflake, postgres, redis.

- Multi-backend ML workflow, data and workflow orchestrator — ML Infrastructure · Noodle.ai
  - Built a generic internal workflow orchestration platform with pluggable backends (airflow, dagster) for feature engineering, triggering and tracking ml workloads (within kubeflow environment) and data orchestration and stitching together workflows consisting of -
    dataset registration, feature jobs' orchestration, ml experimentation, experiment promotion to inference, re-training and analytics pipeline for dashboards showing the performance, error metrics, forecast metrics and feature contributions to the forecast.
  - Designed workloads on this orchestration platform for feature to inference tracking and explainability, possibility of use as ml workflow orchestrator in production for both statistical and neural models with varying resource requirements in a kubernetes environment.

- Infrastructure for user-facing real-time analytics and system observability — Unified Support & Data platform · Zomato
  - Built service components and infrastructure for user-facing real-time analytics to support business observability - Channel Health - capacity and ticket states, tickets inflow/outflow, Support session states, Customer satisfaction, agent handling time metrics, bot resolution proportion; System metrics observability - assignment and session resolution time metrics, assignment states, average messages in a session thread, agent productivity and presence observability and login-hours telemetry - all aggregable at vendor, site, teamlead, channel, agent level.
  - Time-series queries on up to 3 months of data with window approximation to nearest minute or 5th minute on use case and throughput basis with throughput limit of ~0.2 k per second time series queries served with >99.8% success rate under limit of 20-30 ms server time on case by case basis in a tenant-level isolated shared infrastructure.
  - Allowed queries in form of yaml contracts with timeout, max throughput per minute, result caching duration, fields to be aggregated on, filters allowed as configuration for each query and raw queries support for allowing queries on a shared tenant-isolated Apache Pinot cluster in zomato infrastructure [12 nodes of 128GB/1.9TB each] with intention to serve all segments from memory with backing page cache for segment-caching and deep storage of segments on s3. As well as, table onboarding in form of event ingestion contract with protobuf to share pipeline for pinot and hive.
  - Stress-testing infrastructure - ingestion and querying - ingestion pipeline with kafka and real-time pinot tables up to 140k events per minute, querying up to 160k rpm i.e. 2.6 k rps aggregate queries for a 90 day time window served under a limit of 100 ms server time with >98% success rate.

  Predominant scope - real time analytics infrastructure, query engine, system architecture.
  Technologies used - go, grpc, pinot, flink, presto, hive metastore, mysql, mongodb, redis, kafka, aws, k8s, sqs, mqtt, cassandra.

- Custom metrics and reporting workflow builder — Unified Support & Data platform · Zomato
  - Built service components for report/metric builder and preview, report generation and pinning custom metrics on basis of ticket fields both for user defined (custom to each onboarded business on unified support platform) and system fields powered by trino/presto-sql as query execution engine above hive metastore based s3 data warehouse (allowing async report generation queries up to an year of data) and on apache pinot (up to 1 month data).
  - Allowing different timeouts for requests generated from preview on query builder and using trino/presto-sql to do multi-stage join queries on apache pinot for previews at a low throughput expectation and higher timeout limit.
  - Use cases covered - ability to generate reports with filters and aggregates on ticket's current state, ticket's state transitions with, declarative UI based filters & aggregates to build workflow with preview and allowing pinning metric query results on a custom page on dashboard.

- AI Infrastructure & Self-service ML Platform for customer and vertical specific model orchestration and serving — AI Infrastructure / Platform · Unbxd Inc
  - Built platform serving 220+ active merchants with different reference model architectures each for entity extraction, item similarity, search personalization each and training models on-demand specific to each merchant.
  - Resource management, scheduling and orchestration, in a kubernetes environment, for computing data dependencies, data transforms, train/test dataset generation, distributed model training based on allreduce, model validation and orchestrating model version management and automated serving pod deployment post successful run of a model's training pipeline, validation of new serving pod's availability and traffic redirection. Multi-region deployments of this cluster setup serves for model serving and batch workflows for more than 350+ sites.
  - Implemented an api service to allow creating deployments (i.e., creating a recurring run in a kubeflow experiment) for a particular algorithm (a dynamic pipeline i.e., runtime conditional dag of components depending on ancestor outputs). The system is serving more than 700 (without A/B versioning) model deployments and 12-15 batch workflows for all sites as site-wise segregated granular workflows.
  - Config store with dgraph as backend for creating a deployment (recurring algorithm run) and adding deployment configuration (i.e. algorithm inputs, resource template (cpu units, memory, gpu allocation), input and output directories on data lake and s3/gcs corresponding to a recurring run on the platform).
  - CI of model pipelines components and validation of pipeline run in CI environment - categorized under transform, train, validation, test - to validate pipeline run on test site/merchant's catalog and clickstream and ensure model pipeline is completed and serving pod's api server can come up and serve the output model in the CI environment.
  - Adding support for cross-site/merchant vertical specific training for personalization & entity extraction, storing configuration and deployments for a product vertical and orchestrating cross-site vertical clickstream pipeline segregation and recurring training pipelines for top verticals identified under unbxd as a platform.
  - Orchestrating katib experiments for autoML based hyper parameter selection given a set of tuning objectives.
  - Orchestrated existing experimented models for Entity Extraction onto kubeflow pipelines and batch inferences for catalog-specific named entity recognition. Model selection allowed on a platform level for conditional random forests and DNN implementations based on different reference LSTM architectures with bi-lstm, lstm with char cnn, bi-lstm with crf, lstm with char cnn and/or crf (with/wo highway and intent layers).
  - Moved some existing models (with production facing workload mainly for - Query category prediction based on a Siamese network model, Learning to Rank model based on XGB ranker), cross-site/merchant spark transformations i.e., batch spark workflows related to recommendations and reporting, to kubeflow with recurring runs for merchants with standalone spark on segregated clickstream data.

  Predominant scope - ml platform engineering, autoML, ml infrastructure, applied ml, learning to rank, system architecture, distributed systems, ml serving infra.
  Technologies used - go, python, dgraph, redis, kubeflow, katib, argo workflows, katib, kf-serving, openmpi, horovod, spark, keras.

- Personalized search with re-ranking of products and personalized recommendations — Applied AI · Unbxd Inc
  - Experimentation of personalized recommendations with Abacus AI and aws Personalize and self-deployed Matrix factorization (co-occurrence matrix, svd mf, truncated svd mf) based unsupervised models for generating item-item similarity recommendations.
  - Further experimentation with online recommendations with DNN architectures - neural cf (with/wo metadata), neural mf, youtube dnn [1], sequential deep matching [2], deep structured semantic model [3], factorization machines [4] and personalization models used at netcore.Trained models on-demand specific to each merchant catalog data, with youtube dnn as default and further serving personalized search i.e., serving personalized recommendation/re-ranking to customers of each merchant, serving inference traffic at 1-1.2k rps at ingress level.
  - Devised common performing tuning objectives for above architectures for HPO of model with autoML based hyperparameter selection and orchestrating dynamic pipelines for the item similarity (more like this, viewed, bought) and user recommendations & search re-ranking.

    Predominant scope - applied ml, dnn for nlp, ml infrastructure, ml systems - ranking and recommendation systems, search infrastructure, system architecture.
    Technologies used - go, dgraph, redis, python, kubeflow, katib, kf-serving, flask, k8s, aws emr, hive, tf, keras, spark.

- Moving search and indexing services to a multi region - multi tenant infrastructure with isolated indexing — Search & Indexing Infrastructure · Unbxd Inc
  - Worked with a team on a new service managing a search query's life cycle - calling downstream services for pre-search operations - query analysis, synonyms, term boost factors, boosting products, category prediction and post-search operations - reranking search result pages.
    Contributed more towards data platform and model serving integration layer through a new downstream service, dalek, which manages - product boosting, category prediction through batch infer-ed data on redis and and towards entity extraction batch infer-ed data integration on analyzer server, asterix, serving non-cached search throughput up to 80-95 k rpm.
  - Search rerouting from a service, wingman, to aws and gcp regions based on site config fetched from a separate service, that is synced among regions, or to disaster recovery region based on configuration.
  - Orchestrating workflows for isolated indexing on argo workflows with indexing environment solr and post-indexing replication of solr core to primary region shards and replicas as well as disaster recovery region.

    Predominant scope - search infrastructure, distributed systems, multiplexing, dataflow programming, federated search.
    Technologies used - go, go-kit, goflow, java, pippo, solr, nats, postgres, mongodb, dgraph, redis, storm, argo workflows, airbyte, conceptnet, big query, spark, grpc, envoy, k8s, helm.

- Taxonomy for search configuration for indexing and query analyzer and custom boosting on query terms — Merchandising · Unbxd Inc
  - Deployed concept-net backed by postgres and extended the ingestion and catalog pre-processing pipeline to mark, with the help of concept-net graph, taxonomy (concepts, keywords, phrases, synonyms, hypernyms, hyponyms) from catalog in indexer's analyzer configuration and query analyzer configuration to add query synonyms according to selection made by config admin on unbxd merchandising console and mark those selected concepts on concept-net with merchant edge label.

- Re-architecting serviceability infrastructure — Serviceability infrastructure · Zomato
  - Worked on a system, serviceability aggregator, to keep available serviceability levels as a bitmap of serviceability of logistics partner, restaurant, their contract/locality contract timings and rider availability. This serviceability system is the final source of truth for restaurant visibility, aggregating the factors from multiple internal systems including dynamic serviceability.

- Dynamic Serviceability of merchants with point in-time geo-spatial serviceability data — Location & Demand Manager / Experimentation · Zomato
  - Built service and infrastructure components to enable dynamic serviceability of merchants on the platform through geospatial indexing, adding over to existing experimentation with dynamic logistics cost and surge fees based on a demand-elasticity model along with demand forecasts. Merchant serviceability and logistics cost together acts as levers to optimize the utilization of expected total rider available time and provide ability to throttle the demand.
  - With the system, areas where a restaurant can deliver to are governed at runtime at a granularity of level 15 s2 cells equivalent to approx 0.0625 km^2 of square area. There are roughly 2.75 L unique level 15 cells in India where at least a restaurant exists and the system considers a candidate list of more than 7k user cells around the restaurant within 11 Kms of aerial distance i.e. ~ 2 billion source to destination pairs.
  - A rule engine determines, depending on day of week, time slot, user price sensitivity and user behavior, till what road distance or eta should a restaurant service at a particular time with ability to add rules on ttl and priority basis.
    Migrated data for more than 70+ cities, with cell to cell road distances and eta estimates fetched from gmaps api and OSRM, on dynamodb for using rules on road distance or eta as the decision parameter.
    Also added ability to add custom pick/drop cell offsets and rider actual delivery time and drop cell wait time on payload in migrated data on which rules can be formed and linked the system with rain surge automation.
  - The system acts as a delivery promise engine which, provided with a candidate list of restaurants, determines whether the user location would be serviced for each candidate at that point of time. It preprocesses the serviceability data for each restaurant cell for each user segmentation bucket for T and T+1 hours' data. This dataset is stored in a partitioned way on redis and serves a reverse lookup of the user cell, utilizing a custom redis operation and proxy for redis, for each candidate in roughly 3-4 k candidate list of restaurants at 1.2-1.5 k rps non-cached search requests with a p99 latency within 3-4 ms. On the write side, added multi stage workers for optimized pipelined writes.

    Predominant scope - spatial indexing, distributed computing, geo data serving, erlang distribution in stochastic modelling, search infrastructure, distributed systems.
    Technologies used - go, rds mysql, elasticache redis, dynamodb, kafka, elasticsearch, memcached, groupcache, predixy, grpc, envoy, k8s, eks, kinesis, prometheus.

- Connection and job match scoring based on graph analytical modeling — Connection & job match analytics · HT Media
  - Setup pipeline for analytics to export sub-graphs from transactional graph database on neo4j, to analytics graph on titan db, to compute and sync back preferential vertex metrics for over 20 m nodes.
    Used neo4j-mazerunner to implement Bulk Synchronous Parallel [5] to identify dense connected subgraphs and export them for OLAP workload i.e., computation with spark graph x for edge/vertex labeling and scoring, and mark the computed metrics back on the production database. Worked with analytics to manage batch jobs infrastructure in aws emr environment. This drove a rapid rise of referral usage MoM over 6M of observation.

    Predominant scope - distributed graph processing, system architecture, knowledge graph, search infrastructure.
    Technologies used - node.js, python, java, mysql, mongodb, neo4j, redis, rabbitmq, hapi.js, django/drf, restexpress, celery, neo4j-mazerunner, spark graphx, titan db, hbase, apache giraph, apache tinkerpop.